SHUBHAM PATEL

+91 9624069783 | patelshubham152.ps@gmail.com | LinkedIn
Bengaluru, India – 560001.

EDUCATION

- **Pandit Deendayal Energy University** *August 2018 -June 2022*
  B. Tech - Mechanical Engineering Gandhinagar, India.
  - CPI: 9.81/10. | Silver Medalist – Secured 2nd Rank in Institute.

EXPERIENCE

- **Solize India [ ]** *April 2024 - Present*
  Application Engineer Bengaluru, India.
  - Providing technical guidance to clients across aerospace, defense, automotive, and bearing industries for transmission system modelling and simulation using Romax software.
  - Delivering engineering solutions, ensuring high accuracy in drivetrain performance analysis.
  - Support clients in optimizing, designing, and validating transmission systems for enhanced efficiency and reliability.

- **Matter Motor Works [ ]** *Feb 2023 – April 2024*
  Engineer (R&D) – Powertrain Ahmedabad, India.
  - Responsible for executing system level analysis for Strength, Life and Efficiency of Gears, Shafts, Bearings in Romax.
  - Performing Root cause analysis (RCA) to identify and resolve failures during testing and validation phases.
  - Engineered accurate modelling of 4-speed gearbox in Romax for Matter Aera Motorbike.
  - Worked on calculations, design and development of transmission systems for Electric Two-wheeler.
  - Determining gear ratios, speeds, and clutch selections to optimize performance in two-wheeler applications.
  - Supplier Audits, Visits and Process control for tooled up parts ensuring 100% drawing compliance.
  - Responsible for closing DFM's for Mass Production Drawing Release and Tooling Go-aheads.
  - Performed detailed Tolerance & Stack Up calculations for transmission subsystem and assemblies.
  - Conducted Theoretical calculations for Belt & Toroidal CVT and efficiency estimation on Adams for Toroidal CVT.
  - Developed 2D component drawings and managed the release process for manufacturing.
  - Applied DIN & ISO standards to the design and analysis of gears and splines, ensuring reliability.
  - In-depth Understanding Hobbing, Broaching, Grinding, Manufacturing Process & Heat treatment of Gears.

- **AAAG India [ ]** *Dec 2021 – May 2022*
  Research & Development Intern Ahmedabad, India.
  - Developed firefighting robot design for petrochemical and High heat release fire applications.
  - Analyzed industry requirements and challenges to refine technical specifications for a fire-fighting robot.
  - Engineered a robust tracked system featuring a tensioner and durable rubber steel belt for enhanced traction.
  - Implemented an all-wheel suspension system for stability and agility.
  - Integrated water monitor, drive system, controller in 1600mm x 1200mm base for optimal suitability and agility.
  - Optimized hose connections to enhance towing efficiency and deployment ease for firefighting applications.

- **Edufeat [ ]** *Jan 2021 – June 2022*
  Subject Matter Expert Remote.
  - Delivered live sessions and resolved complex engineering problems in subjects including.
  - Fluid Mechanics | Strength of Materials | Fluid Dynamics | MATLAB | Machine Design | Engineering Mathematics.

PROJECTS

- **Baja SAE India -** Intercollegiate All-Terrain Vehicle (ATV) Engineering Design Competition. *Feb 2019 – Jan 2021*
  Analysis Department Head, Team Czar.
  - Led the Analysis department, performing Thermal, static, fatigue analysis of Critical components on SIEMENS NX.
  - Prepared Jigs, fixtures and carried out assembly and commissioning of the ATV.
  - Conducted RCA and CAPA of the failures encountered during the testing.
  - Contributed to Project Planning, component manufacturing & Testing of ATV.
  - Prepared the of DFMEA, DVP&R sheet for critical components of the vehicle.
  - Led the Design and testing Gear box and Differential system of powertrain.

SKILLS

- **Design:** 2D Drawings, 3D Modelling (Creo, Siemens Nx) | RCA | GD&T | DFM | DFMEA
- **Simulation:** FEA | Romax | Kissoft | Adams
- **Programming & Control:** Python | MATLAB
- **Manufacturing & Process:** Forging | Machining | Hobbing & Shaving | Broaching | Honning | Shot Peening | 3D printing
- **Analytical:** Calculations | Teamwork | Time Management

HONORS AND AWARDS

- **Silver Medal Award** *June 2022*
  Pandit Deendayal Energy University
  - Received this for securing 2nd rank in B. Tech Mechanical Engineering, Batch of 2018.
- **Scholarships** *June 2018-June 2022*
  Educational Department, State of Gujarat.
  - Scholarship amount of Rs 2,00,000 over the course of B. Tech for performance in High School.

CERTIFICATIONS

- Python Data Structures| Control of Mobile Robots | 3D printing Applications - Coursera.
- Essentials for NX Designers – Siemens Centre of Excellence. *Aug 2019-Dec 2019*